



AUTOXR: REAL-TIME AR/VR VEHICLE CUSTOMIZATION

Mohammed Tanvir Rasheed, Ramana Prabhu
Sana, Vinay Vuyyuri, Pallavi Yerramilli

Professor: Alex Moehring

MGMT 682 – Digital Product Design



APPLE VISION PRO

Table of Contents

1. Product and Customer Need
2. Problem, End Benefit, and Target Segment
3. Data and Methods
4. Choice-Based Conjoint (CBC)
5. Results and Discussion
6. Simulations
7. Conclusion & Recommendations
8. Limitations



Problem Statement

Market Needs & Value Proposition

- Vehicle buyers struggle to visualize colors, parts, and accessories accurately before purchase.
- This leads to uncertainty, long decision cycles, and costly rework for dealers and custom shops.
- AutoXR solves this by projecting full-scale, true-to-life AR customization directly onto the actual vehicle.
- Survey (N=40):
 - **72.5%** prefer AR over standard previews
 - **65%** spend **3+ hours** customizing
 - **82.5%** choose price ranges **< \$2,000**
 - AutoXR creates value through realism, faster decisions, and fewer ordering mistakes.

Insights, Methods & Pricing Logic

Data, Methods & EVC Market Anchor

Plans (Importance Score)	XR Realism (Importance Score)	Where you can use it (Importance Score)	Saved configurations (Importance Score)	Collaboration features (Importance Score)	Monthly price (Importance Score)
0.118663548	0.044594909	0.190024727	0.077225041	0.091433885	0.47805789
0.093898516	0.192345721	0.041596359	0.117605628	0.203665599	0.350888176
0.1069836	0.078506837	0.183578219	0.076824547	0.071221046	0.482885751
0.085958591	0.136475105	0.131155103	0.100244426	0.085945371	0.460221404
0.134883879	0.161686129	0.05226401	0.081575833	0.207149756	0.362440393
0.066179747	0.204851681	0.021645655	0.143855411	0.314092579	0.249374926
0.109032527	0.04835253	0.175323631	0.069808666	0.096389483	0.501093162

Data & Methods:

- 40 survey responses on customization habits, time spent, AR usefulness, and price perception.
- Analysis combined descriptive statistics, willingness-to-pay indicators, and competitor benchmarks.

EVC Market Anchor:

- Competitor pricing:
 - Unreal: **\$1,850/year**
 - Unity Forma: **Enterprise (5–6 figures)**
 - ZeroLight: **\$100k+/year**
- Survey anchors:
 - **82.5%** expect costs < \$2k
 - Supports AutoXR’s target pricing: **\$499–\$1,499/month** for dealers.

Differentiators & Conjoint Attributes

Why AutoXR Wins & How Attributes Were Chosen

Differentiators

- True-scale AR on the real vehicle
- High-fidelity, lighting-accurate visuals
- Dealer-ready workflow & single price bundle
- Reduces rework + uncertainty

Survey Signal:

Users said current tools feel “**uncertain**” and “**not accurate**” → supports realism + AR alignment. bundle attributes.

What Users Want

- Realistic colors & environment matching
- Ability to see mods on *their own* car
- Simple, fast configuration flow

Survey Signal:

Open-ended feedback mentioned **color accuracy, lighting, fit confidence,** and desire to **avoid mistakes.**

Attributes

- Realism level & lighting accuracy
- AR alignment on actual vehicle
- Catalog depth
- Speed & ease of use
- Bundle clarity (single priced package)

Survey Signal:

Users asked for **more modification options**, fewer steps, and **clearer final pricing** → justifies catalog, speed, and bundle attributes.

AR Training for Automotive Mechanics

Conjoint Insights and Value Proposition

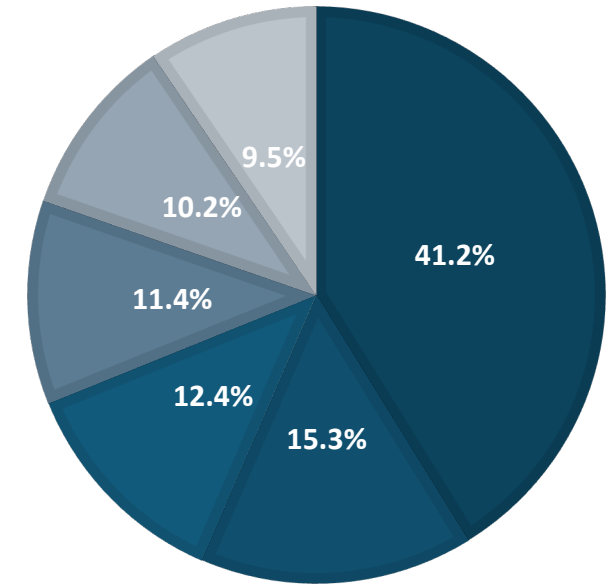
- **Product:** AutoXR is an XR training and demo solution for automotive customers.
- **Experience:** Uses immersive XR to show vehicle features and guide complex tasks.
- **Goal:**
 - Help users understand advanced features faster.
 - Reduce confusion during the sales process and after delivery.
- **Study:**
 - Online choice-based **conjoint survey**.
 - Attributes: Plans, XR realism, where you can use it, saved configurations, collaboration features, monthly price.
 - We use the results to define **target segments** and a **pricing strategy**.



What Drives Choice Preferences - Attribute Importance

- Our conjoint results show the following **importance scores**:
 - Monthly price: 41.2%
 - Collaboration features: 15.3%
 - XR Realism (content quality): 12.4%
 - Where you can use it (showroom vs showroom + home): 11.4%
 - Plans (feature bundle): 10.2%
 - Saved configurations: 9.5%
- **Interpretation:**
 - Price is the single largest driver of choice, which fits a recurring monthly subscription.
 - Non-price features still matter a lot: collaboration + XR realism + access location together account for almost 40% of decisions.
 - Customers value useful collaboration tools and high-quality XR realism, not just a cheap plan.

ATTRIBUTE IMPORTANCE SPLIT



- Monthly price
- Collaboration features
- XR realism (content quality)
- where you can use it (showroom vs showroom+home)
- plans (feature bundle)
- saved configurations

Trouble with Current Solutions (Content + Access)

Current Solutions and Unmet Needs

XR Realism:

- Level 3 - High detail + reflections and shadows: +37.42 utils
- Level 2 - High detail + lighting realism: -36.88 utils
- Level 1 - Basic 3D overlay level: -0.54 utils

Interpretation:

- Only the **highest-realism content** delivers clearly positive value.
- Mid-level realism and basic overlays sit at or below zero, behaving like “good enough but not exciting” status quo.

Where you can use it:

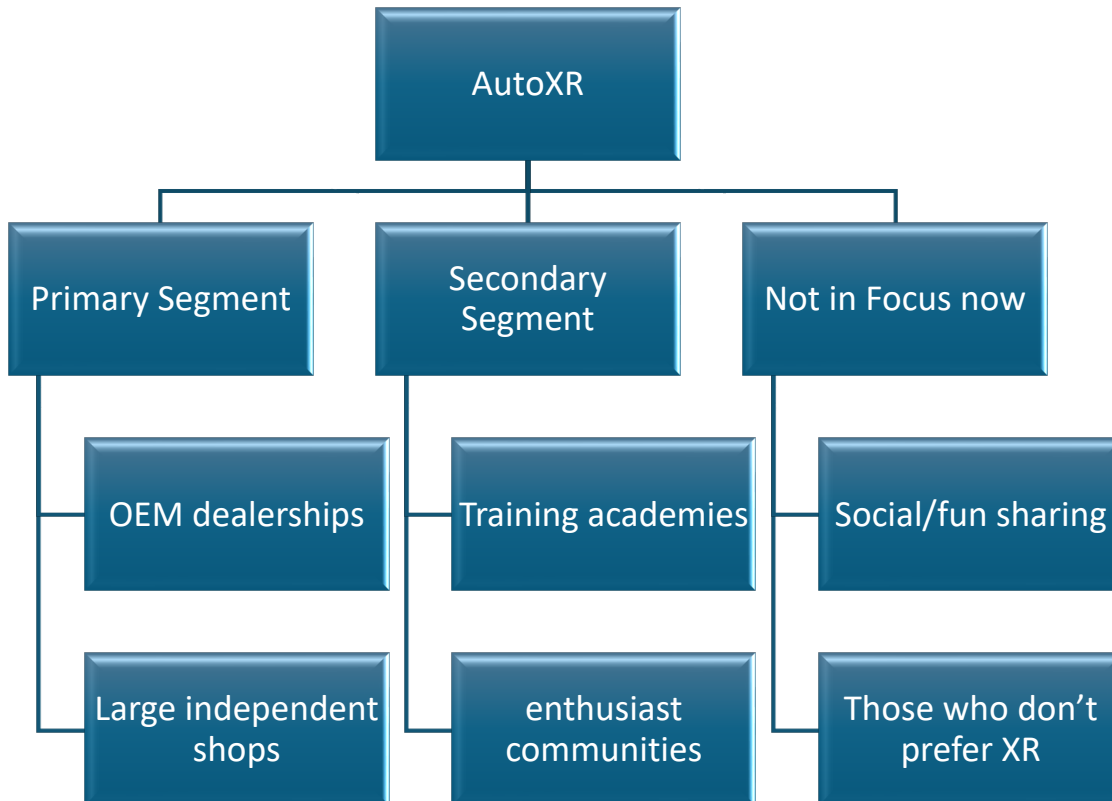
- **Showroom + at home: +32.24 utils**
- **Showroom only: -32.24 utils**

Interpretation:

- Experiences locked to a **single showroom location** have strong negative utility.
- Users clearly prefer XR that **follows them home** and extends the value of the demo.

Target Users and Segmentation

Target Users - Who AutoXR Serves First



Collaboration features:

- **Dealer annotations:** +20.30 utils
- **Friends/family sharing:** +6.70 utils
- **No collaboration:** -27.00 utils

Interpretation:

- Customers **strongly dislike having no collaboration.**
- They value **dealer annotations** much more than simple social sharing, which fits a **professional, sales-oriented** workflow rather than casual hobby use.

Primary target segment:

- **OEM dealerships and larger independent shops** that:
 - Need clear feature explanations and delivery walkthroughs.
 - Dealer-annotation feature matches their sales and service workflow.

Secondary segment (later): Training academies or enthusiast communities where “share with friends/family” is more important.

Not in Focus now: Social / fun sharing only (friends & family), Users who prefer **no XR solution**

Utilities and Main Drivers

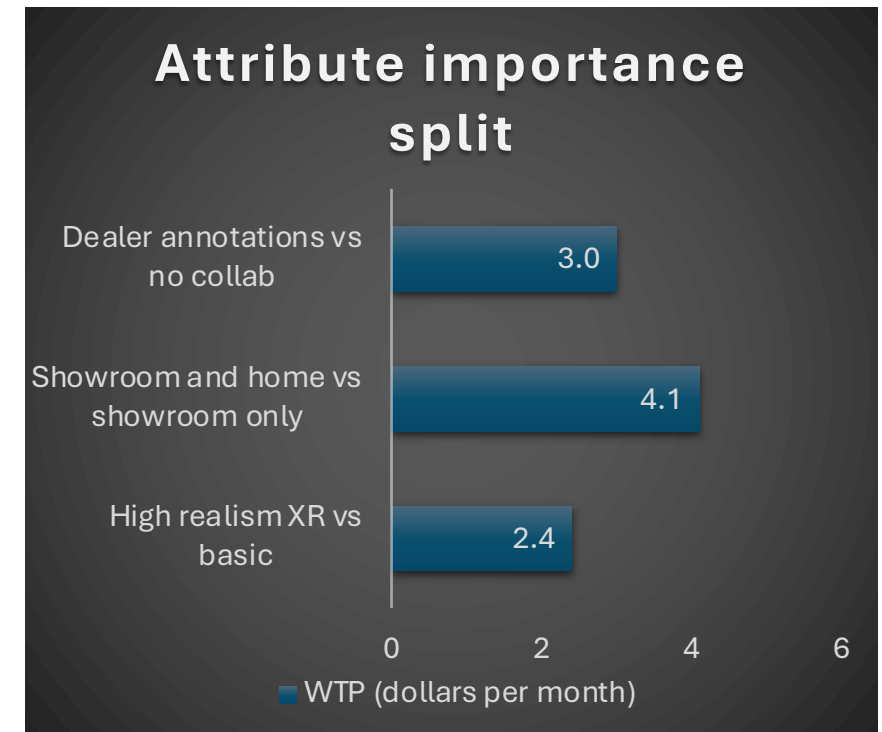
Part Worth Utilities – Best vs
Worst Profiles

- **XR realism (content):**
 - Level 3 (high detail + reflections): **+37.42 utils**,
 - Level 2 (lighting only): **-36.88 utils** or
 - Level 1 (basic overlay): **-0.54 utils**
 - → Only **full realism (Level 3)** is clearly preferred.
- **Where you can use it (Access):**
 - Showroom + at home: **+32.24 utils** or
 - Showroom only: **-32.24 utils** → Strong penalty for **showroom-only** experiences.
- **Collaboration:**
 - Dealer annotations: **+20.3 utils** Friends/family sharing: **+6.7 utils** or None: **-27.0 utils** → Customers want **professional guidance**, not no-collab.
- **Price:** \$5/month: **+135.8 utils**, \$10/month: **-38.2 utils** or \$20/month: **-97.6 utils** → Clear **price sensitivity** across the range.
- **Overall:**
 - **Best profile:** Level 3 realism, showroom + home, dealer annotations, lower price.
 - **Worst profile:** Mid/low realism, showroom-only, no collaboration, high price.

Willingness to Pay for Key Features

Willingness to Pay – Using the Price Attribute

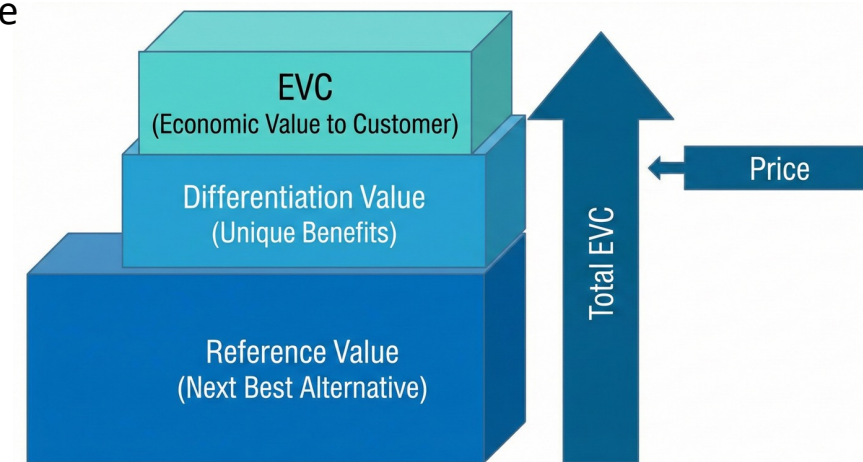
- From the **price utilities** (5 dollars vs 20 dollars per month), the model implies:
 - One util is worth about **\$0.06 per month**
- **Extra willingness to pay per month compared to a basic option:**
 - **High realism XR (Level 3 instead of Level 1)**
 - Utility difference is about 38 utils. This is worth about **\$2.4 more per month**
 - **Use AutoXR in the showroom and at home (instead of showroom only)**
 - Utility difference is about 64.5 utils. This is worth about **\$4.1 more per month**
 - **Dealer annotations (instead of no collaboration)**
 - Utility difference is about 47.3 utils. This is worth about **\$3.0 more per month**
- **Interpretation (link to lecture concepts):**
 - These numbers show the **monetary value of each feature** on the demand curve.
 - Customers are willing to pay several dollars more per month for:
 - Better content, more usage occasions, better guidance.
 - This supports a **two-tier strategy**: a basic plan and a premium plan that captures this extra value.



EVC Analysis

Competitive Landscape & Reference Value

- **Reference Anchor:**
 - Unreal Automotive Configurator (\$1,850/yr).
 - High fidelity, but lacks dealer workflows and full-scale AR integration.
- **Market Ceiling:** Enterprise solutions like Unity Forma/ZeroLight cost more (\$100k+).
- **AutoXR Opportunity:**
 - Fills the "Mid-Market" gap
 - More capable than raw tools (Unreal), more accessible than enterprise engines.
- **Survey Validation:**
 - 72.5% found reference prices "about right" or "too low,"
 - Users value realism over budget tools.
- **Conclusion:** AutoXR's base utility exceeds the \$1,850/yr benchmark.





EVC Analysis

Value Drivers & Positioning

- **AR Superiority:** 72.5% rate AR better than standard previews; AutoXR adds unique full-scale alignment on physical cars
- **Operational Efficiency:** 65% of users spend 3+ hours customizing. AutoXR drastically accelerates this, saving dealer labor costs..
- **Visual Trust:** Open-ended comments emphasize the critical need for accurate visuals. AutoXR mitigates the customer remorse.
- **Final Pricing:** The price premium over Unreal (~\$154/mo vs. AutoXR's ~\$500+) is justified because AutoXR is sold as a **sales platform**

Choice-Based Conjoint (CBC) Analysis

Why CBC?

- Find Trade-offs between features and cost
- Calculate MVP
- Calculate Price Elasticity and WTP

Key Attributes:

- **XR Realism:** *Basic Overlay* vs. *High-Fidelity Reflections*
- **Usage Location:** *Showroom Only* vs. *+ At Home*
- **Save Configuration:** *3 Setups* vs. *10 Setups*
- **Monthly Price:** \$5, \$10, \$20

Sampling & Estimation

- Mixed Population of Digital and Traditional Users, mainly students.
- Utility Scores to WTP

Simulations Overview

Goal: Converting conjoint utilities into expected adoption at different prices

Approach:

- Use CBC results to simulate choices between:
 - Premium bundle (Plan B, Level 3 realism, showroom + home, 10 saves, dealer annotations)
 - Basic bundle (Plan A, Level 1 realism, showroom only, 3 saves, no collaboration, \$10/month)
 - None option (“I would not choose any plan”)
- Include None in all simulations to reflect real opt-out behavior

Calibration:

- Respondents chose None in 18.3% of CBC tasks
- We calibrate the simulations so model “None” share matches 18.3% at the \$10 reference point



Take-up vs Price

Simulation setup:

- Premium price varies; Basic is fixed at \$10/month
- Premium bundle includes Plan B, Level 3 realism, showroom + home, 10 saves, dealer annotations

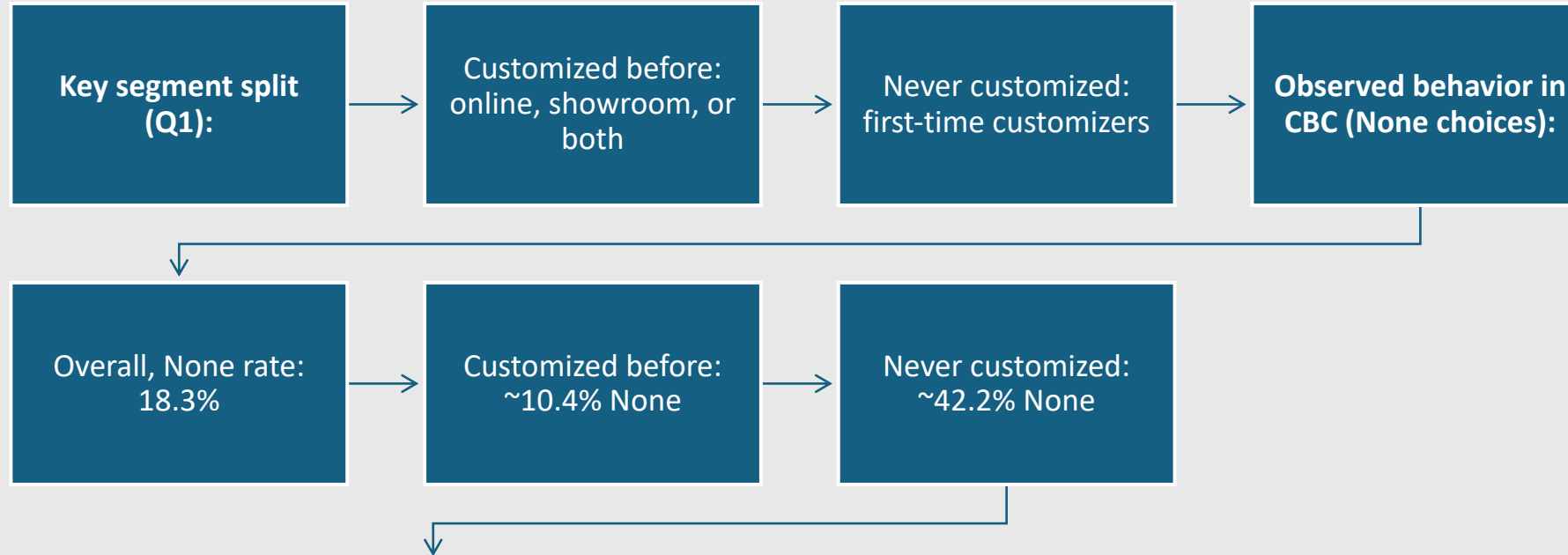
Key patterns from the simulation:

- Premium adoption is strong in the \$14–\$15/month range
 - Take-up remains around 50–58% in this zone
- As price increases toward \$20/month,
 - None share grows sharply
 - Many users drop out rather than accept a high premium price
- Basic plan stays weak because it combines low realism, showroom-only access, and no collaboration, all of which have negative utilities

Implication:

- The \$14–\$15/month range is a good starting point for a Premium tier: still attractive, without competing purely on being the cheapest option.

Segment Comparison



Simulation at \$15/month Premium:

- The Premium bundle at \$15/month is strongly attractive for experienced customizers
- First-time customizers are much more likely to choose None, even at the same price
- This segment likely needs:
 - A lighter, simpler trial or entry plan, or
 - A dealer-guided onboarding flow that explains the tool and options

Segment	Premium share	None share
Customized before	66.2%	33.8%
Never customized	23.6%	76.4%

Overall Conclusion



1. Product fit:

AutoXR's value lies in high realism, showroom + home access, and guided collaboration (dealer annotations)



2. Customer segment:

Best near-term fit: OEM dealerships and larger independent shops serving buyers who already customize online or in showrooms



3. Pricing strategy:

Use a tiered structure rather than a single flat price
Position Premium in the \$14–\$15/month range (per end-user equivalent), anchored by EVC and simulations

For dealers, a \$150/location/month subscription is perceived as close to “about right” on average



4. Adoption risk:

Comfort (motion sickness) and privacy are key concerns that must be actively managed

Recommended Configuration & Price

Recommended Premium configuration (v1):

Plan: Plan B
XR realism: Level 3 (high detail + reflections)
Where to use: Showroom and home
Saved configurations: Up to 10 setups
Collaboration: Dealer annotations

Why this configuration:

Aligns with highest utilities on realism, access, and collaboration
Matches survey attitudes: realism and confidence (Q5, Q6, Q8) and desire to use AutoXR (Q11)

Recommended pricing:

Basic: \$10/month (low realism, showroom-only, minimal collaboration)
Premium: \$14–\$15/month
≈ 70–75% of calculated EVC (~\$19.6/month)
Supported by simulation take-up and willingness-to-pay
Dealer package: AutoXR Pro at \$150/location/month with ROI justification (upsell, faster explanations, fewer post-sale questions)

Limitations

Sample and context:

- n = 40, mostly students and early-career professionals
- Not fully representative of all dealership customers or markets

Stated preference vs real behavior:

- CBC choices may not perfectly match real-world decisions, especially at higher prices

Comfort and motion sickness:

- Most frequent open-ended concern; real-world adoption may be lower if XR comfort is not addressed

Pricing model skepticism:

- Some respondents questioned why individuals would pay a subscription
- Supports a B2B focus on dealers and OEMs with clear ROI framing

Privacy:

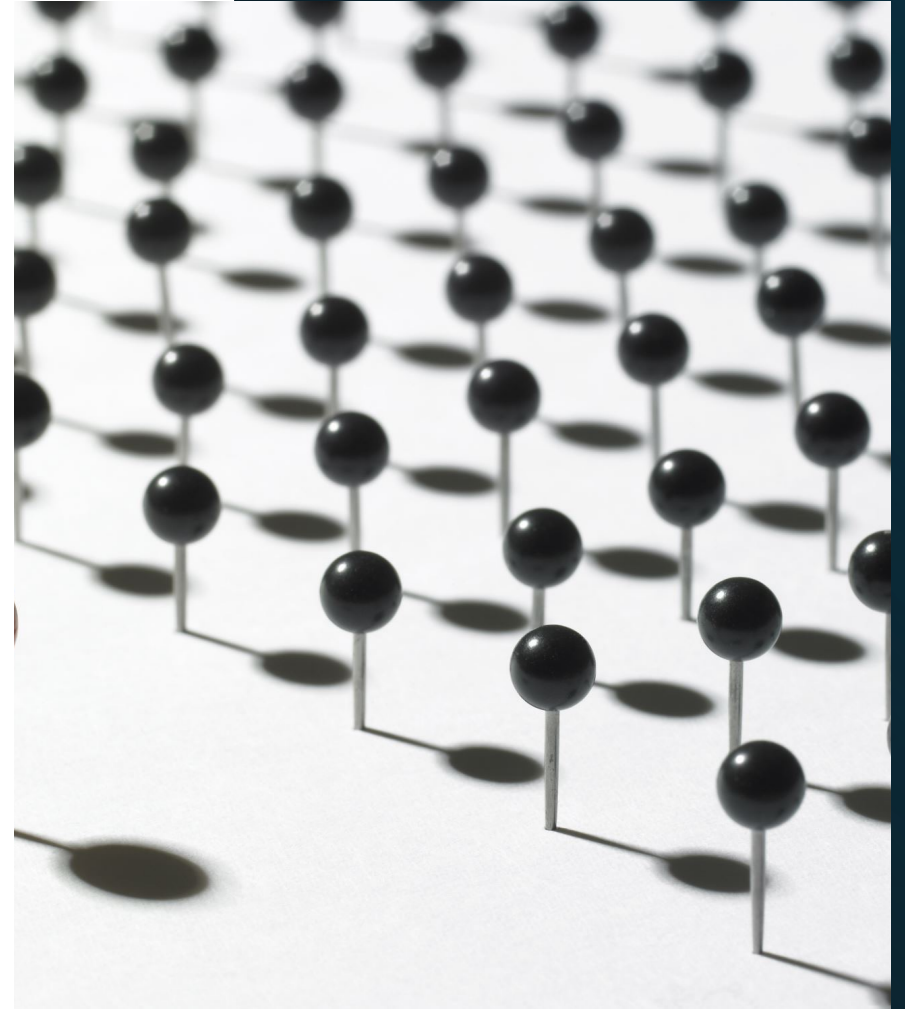
- Several respondents mentioned privacy; data capture on or around vehicles must be designed carefully

Simulation scope:

- Simulations assume static bundles and do not fully capture hardware availability, dealer training, or in-dealership experience

References

- [1] Orme, B. K. (2014). Getting started with conjoint analysis: Strategies for product design and pricing (3rd ed.). Madison, WI: Research Publishers LLC.
- [2] Sawtooth Software. CBC/HB: Software for hierarchical Bayes estimation of individual-level utilities. Retrieved December 9, 2025, from Sawtooth Software website.
- [3] Louviere, J. J., Hensher, D. A., & Swait, J. D. (2000). Stated choice methods: Analysis and application. Cambridge, UK: Cambridge University Press.
- [4] Epic Games. Unreal Engine for automotive configurators. Retrieved December 9, 2025, from the Unreal Engine automotive solutions page.
- [5] Unity Technologies. Unity Forma: Real-time 3D marketing for automotive products [Software]. Retrieved December 9, 2025, from the Unity Forma product page.
- [6] ZeroLight Ltd. ZeroLight: Real-time 3D visualization and car configurator platform. Retrieved December 9, 2025, from the ZeroLight solutions overview page.



Any Questions?

Thank you

